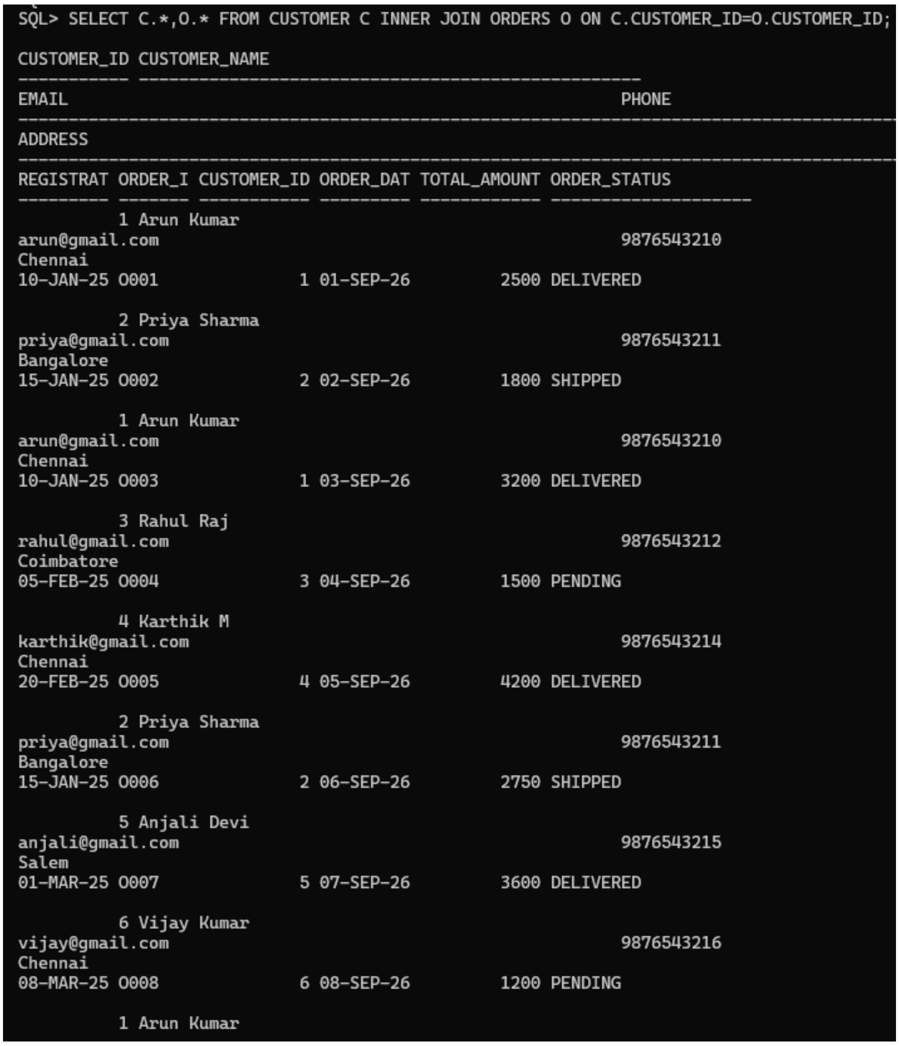
WEEK 8 : Database Relationship Analysis using Joins

Scenario:

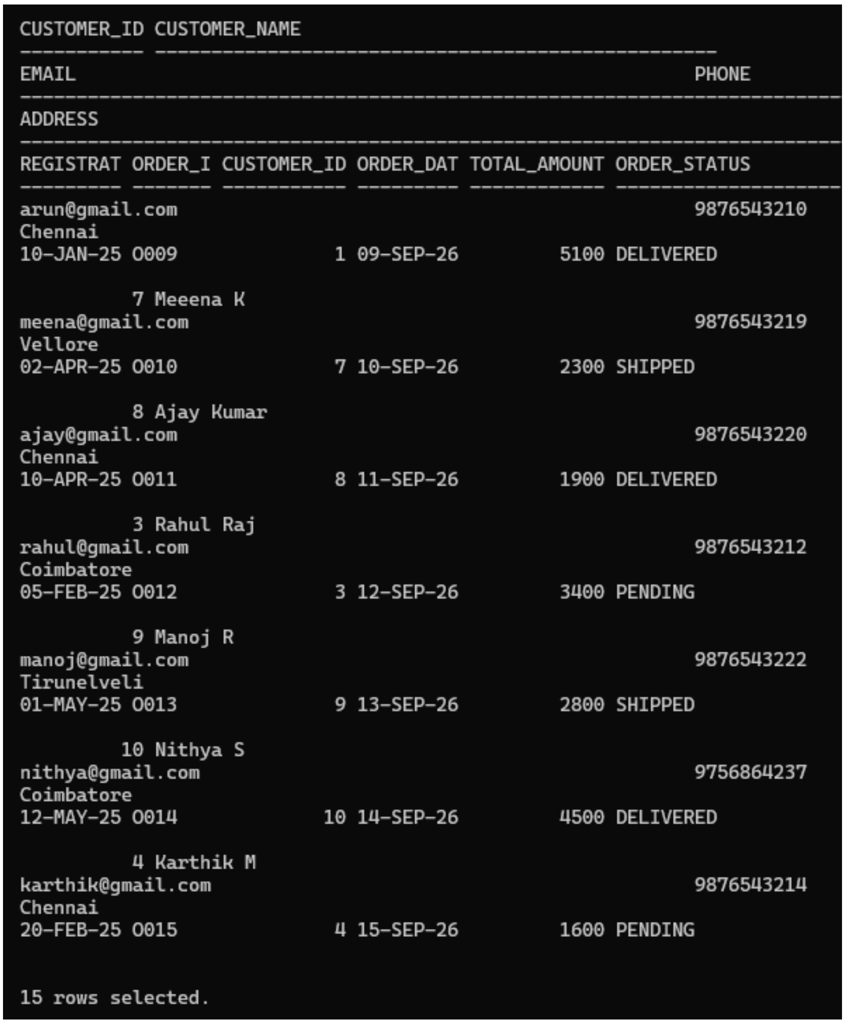
The e-commerce company stores customer, product, order, and payment information in separate tables. The management team wants to combine these related tables to understand customer purchases, order details, and payment status.

1. **Implement INNER JOIN**

   Display customer details along with their orders.

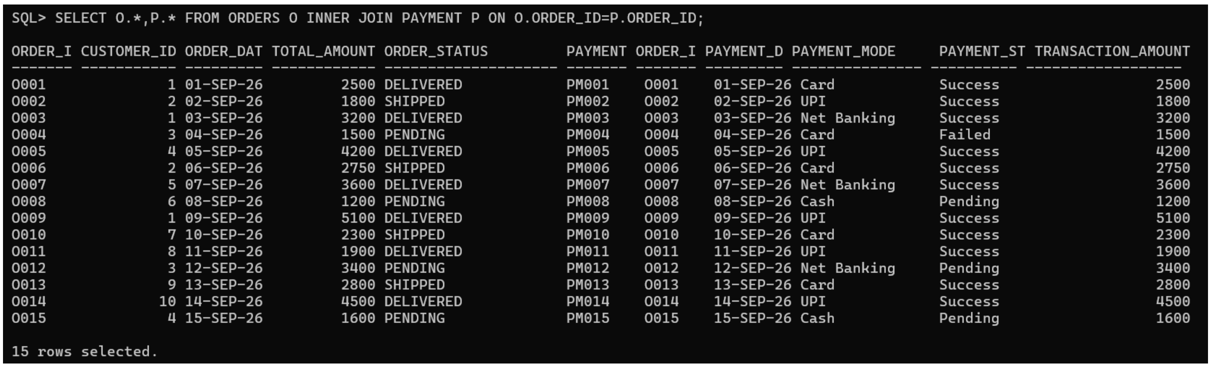


```
SQL> SELECT C.*,O.* FROM CUSTOMER C INNER JOIN ORDERS O ON C.CUSTOMER_ID=O.CUSTOMER_ID;

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL                                                       PHONE
---------------------------------------------------------------------------------------
ADDRESS
---------------------------------------------------------------------------------------
REGISTRAT ORDER_I CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
--------- ------- ----------- --------- ------------ --------------------
          1 Arun Kumar
arun@gmail.com                                              9876543210
Chennai
10-JAN-25 O001              1 01-SEP-26         2500 DELIVERED

          2 Priya Sharma
priya@gmail.com                                             9876543211
Bangalore
15-JAN-25 O002              2 02-SEP-26         1800 SHIPPED

          1 Arun Kumar
arun@gmail.com                                              9876543210
Chennai
10-JAN-25 O003              1 03-SEP-26         3200 DELIVERED

          3 Rahul Raj
rahul@gmail.com                                             9876543212
Coimbatore
05-FEB-25 O004              3 04-SEP-26         1500 PENDING

          4 Karthik M
karthik@gmail.com                                           9876543214
Chennai
20-FEB-25 O005              4 05-SEP-26         4200 DELIVERED

          2 Priya Sharma
priya@gmail.com                                             9876543211
Bangalore
15-JAN-25 O006              2 06-SEP-26         2750 SHIPPED

          5 Anjali Devi
anjali@gmail.com                                            9876543215
Salem
01-MAR-25 O007              5 07-SEP-26         3600 DELIVERED

          6 Vijay Kumar
vijay@gmail.com                                             9876543216
Chennai
08-MAR-25 O008              6 08-SEP-26         1200 PENDING

          1 Arun Kumar
```



```
CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL                                                       PHONE
--------------------------------------------------------------------------
ADDRESS
--------------------------------------------------------------------------
REGISTRAT ORDER_I CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
--------- ------- ----------- --------- ------------ --------------------
arun@gmail.com                                              9876543210
Chennai
10-JAN-25 O009              1 09-SEP-26         5100 DELIVERED

          7 Meeena K
meena@gmail.com                                             9876543219
Vellore
02-APR-25 O010              7 10-SEP-26         2300 SHIPPED

          8 Ajay Kumar
ajay@gmail.com                                              9876543220
Chennai
10-APR-25 O011              8 11-SEP-26         1900 DELIVERED

          3 Rahul Raj
rahul@gmail.com                                             9876543212
Coimbatore
05-FEB-25 O012              3 12-SEP-26         3400 PENDING

          9 Manoj R
manoj@gmail.com                                             9876543222
Tirunelveli
01-MAY-25 O013              9 13-SEP-26         2800 SHIPPED

         10 Nithya S
nithya@gmail.com                                            9756864237
Coimbatore
12-MAY-25 O014             10 14-SEP-26         4500 DELIVERED

          4 Karthik M
karthik@gmail.com                                           9876543214
Chennai
20-FEB-25 O015              4 15-SEP-26         1600 PENDING


15 rows selected.
```

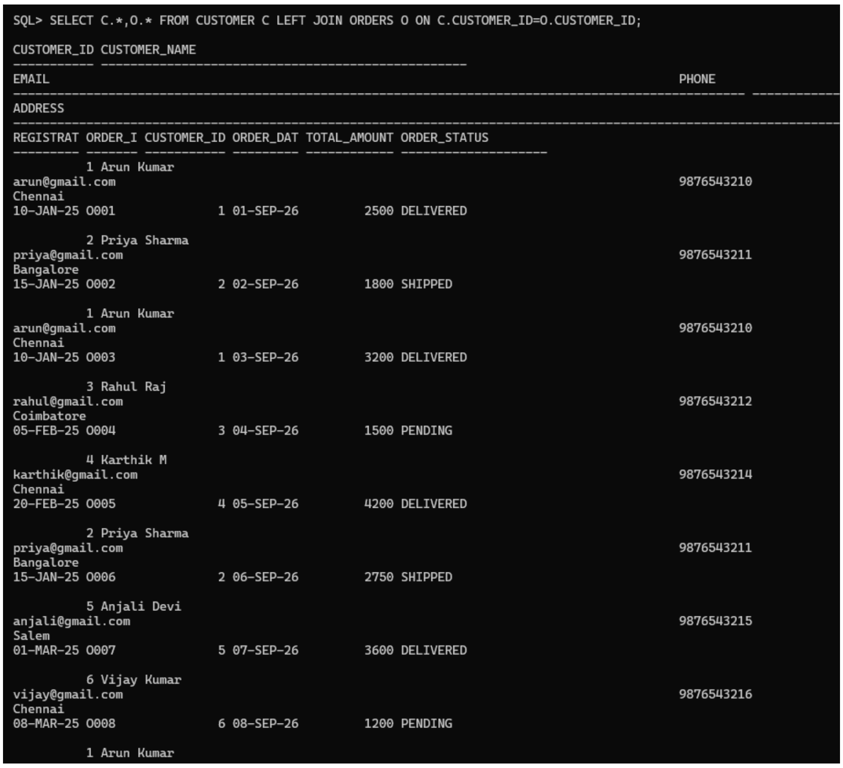
Display order details with payment information.



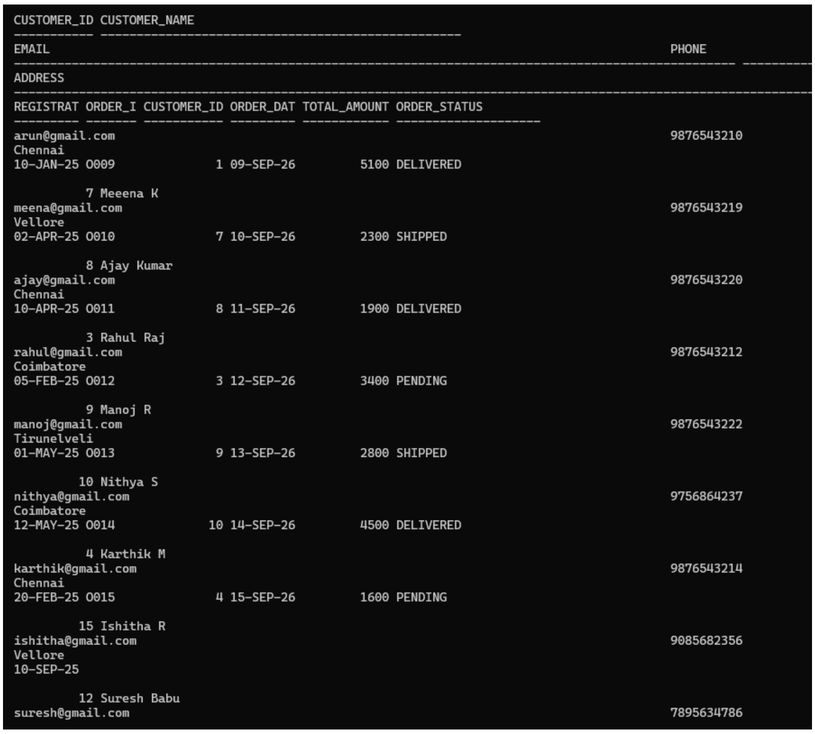
```
SQL> SELECT O.*,P.* FROM ORDERS O INNER JOIN PAYMENT P ON O.ORDER_ID=P.ORDER_ID;

ORDER_I CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS         PAYMENT ORDER_I PAYMENT_D PAYMENT_MODE     PAYMENT_ST TRANSACTION_AMOUNT
------- ----------- --------- ------------ -------------------- ------- ------- --------- --------------- ---------- ------------------
O001              1 01-SEP-26         2500 DELIVERED            PM001    O001    01-SEP-26 Card            Success                  2500
O002              2 02-SEP-26         1800 SHIPPED              PM002    O002    02-SEP-26 UPI             Success                  1800
O003              1 03-SEP-26         3200 DELIVERED            PM003    O003    03-SEP-26 Net Banking     Success                  3200
O004              3 04-SEP-26         1500 PENDING              PM004    O004    04-SEP-26 Card            Failed                   1500
O005              4 05-SEP-26         4200 DELIVERED            PM005    O005    05-SEP-26 UPI             Success                  4200
O006              2 06-SEP-26         2750 SHIPPED              PM006    O006    06-SEP-26 Card            Success                  2750
O007              5 07-SEP-26         3600 DELIVERED            PM007    O007    07-SEP-26 Net Banking     Success                  3600
O008              6 08-SEP-26         1200 PENDING              PM008    O008    08-SEP-26 Cash            Pending                  1200
O009              1 09-SEP-26         5100 DELIVERED            PM009    O009    09-SEP-26 UPI             Success                  5100
O010              7 10-SEP-26         2300 SHIPPED              PM010    O010    10-SEP-26 Card            Success                  2300
O011              8 11-SEP-26         1900 DELIVERED            PM011    O011    11-SEP-26 UPI             Success                  1900
O012              3 12-SEP-26         3400 PENDING              PM012    O012    12-SEP-26 Net Banking     Pending                  3400
O013              9 13-SEP-26         2800 SHIPPED              PM013    O013    13-SEP-26 Card            Success                  2800
O014             10 14-SEP-26         4500 DELIVERED            PM014    O014    14-SEP-26 UPI             Success                  4500
O015              4 15-SEP-26         1600 PENDING              PM015    O015    15-SEP-26 Cash            Pending                  1600

15 rows selected.
```

2. Implement LEFT JOIN

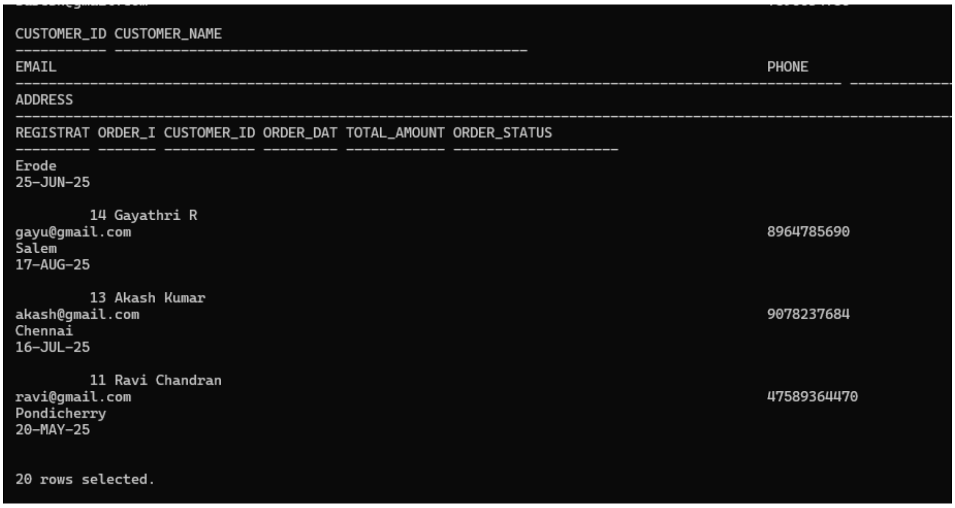
Display all customers including customers who have not placed orders.



```
SQL> SELECT C.*,O.* FROM CUSTOMER C LEFT JOIN ORDERS O ON C.CUSTOMER_ID=O.CUSTOMER_ID;

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL                                                                                      PHONE
------------------------------------------------------------------------------------------ ------------
ADDRESS
------------------------------------------------------------------------------------------------------------
REGISTRAT ORDER_I CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
--------- ------- ----------- --------- ------------ --------------------
          1 Arun Kumar
arun@gmail.com                                                                             9876543210
Chennai
10-JAN-25 0001              1 01-SEP-26         2500 DELIVERED

          2 Priya Sharma
priya@gmail.com                                                                            9876543211
Bangalore
15-JAN-25 0002              2 02-SEP-26         1800 SHIPPED

          1 Arun Kumar
arun@gmail.com                                                                             9876543210
Chennai
10-JAN-25 0003              1 03-SEP-26         3200 DELIVERED

          3 Rahul Raj
rahul@gmail.com                                                                            9876543212
Coimbatore
05-FEB-25 0004              3 04-SEP-26         1500 PENDING

          4 Karthik M
karthik@gmail.com                                                                          9876543214
Chennai
20-FEB-25 0005              4 05-SEP-26         4200 DELIVERED

          2 Priya Sharma
priya@gmail.com                                                                            9876543211
Bangalore
15-JAN-25 0006              2 06-SEP-26         2750 SHIPPED

          5 Anjali Devi
anjali@gmail.com                                                                           9876543215
Salem
01-MAR-25 0007              5 07-SEP-26         3600 DELIVERED

          6 Vijay Kumar
vijay@gmail.com                                                                            9876543216
Chennai
08-MAR-25 0008              6 08-SEP-26         1200 PENDING

          1 Arun Kumar
```

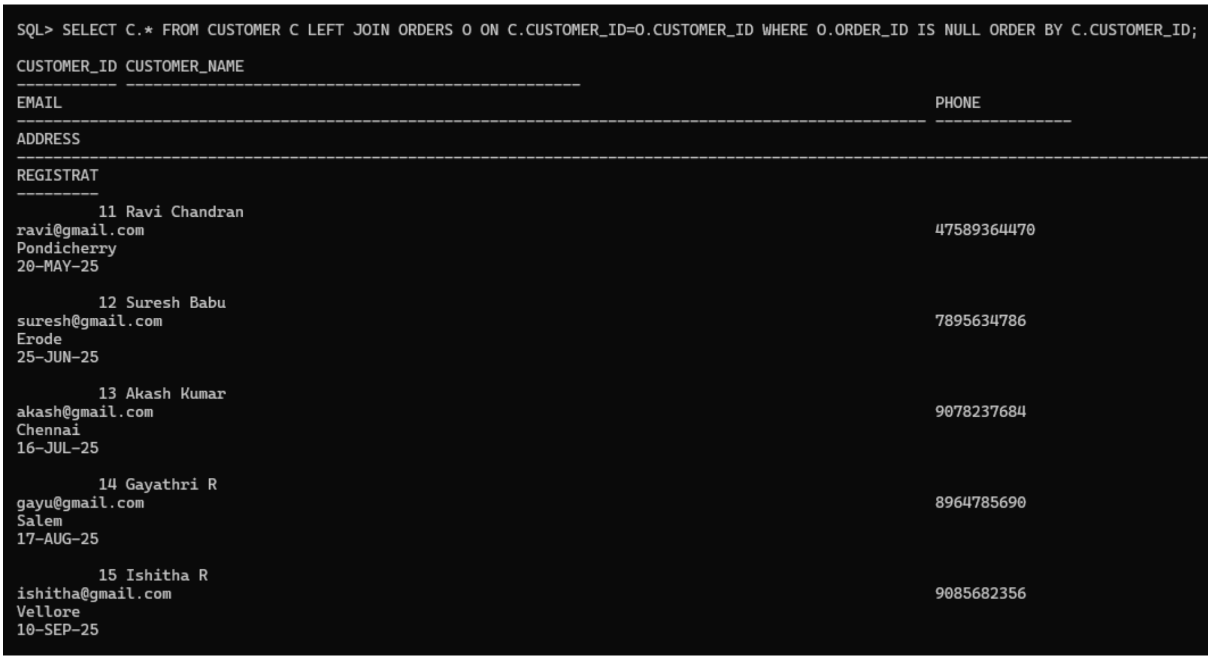


```
CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL                                                                                      PHONE
------------------------------------------------------------------------------------------ ----------
ADDRESS
----------------------------------------------------------------------------------------------------------
REGISTRAT ORDER_I CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
--------- ------- ----------- --------- ------------ --------------------
arun@gmail.com                                                                             9876543210
Chennai
10-JAN-25 0009              1 09-SEP-26         5100 DELIVERED

          7 Meeena K
meena@gmail.com                                                                            9876543219
Vellore
02-APR-25 0010              7 10-SEP-26         2300 SHIPPED

          8 Ajay Kumar
ajay@gmail.com                                                                             9876543220
Chennai
10-APR-25 0011              8 11-SEP-26         1900 DELIVERED

          3 Rahul Raj
rahul@gmail.com                                                                            9876543212
Coimbatore
05-FEB-25 0012              3 12-SEP-26         3400 PENDING

          9 Manoj R
manoj@gmail.com                                                                            9876543222
Tirunelveli
01-MAY-25 0013              9 13-SEP-26         2800 SHIPPED

         10 Nithya S
nithya@gmail.com                                                                           9756864237
Coimbatore
12-MAY-25 0014             10 14-SEP-26         4500 DELIVERED

          4 Karthik M
karthik@gmail.com                                                                          9876543214
Chennai
20-FEB-25 0015              4 15-SEP-26         1600 PENDING

         15 Ishitha R
ishitha@gmail.com                                                                          9085682356
Vellore
10-SEP-25

         12 Suresh Babu
suresh@gmail.com                                                                           7895634786
```



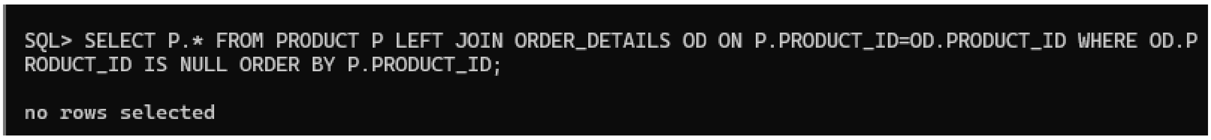
```
CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL                                                                                                PHONE
---------------------------------------------------------------------------------------------------- -------------
ADDRESS
--------------------------------------------------------------------------------------------------------------------
REGISTRAT ORDER_I CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
--------- ------- ----------- --------- ------------ --------------------
Erode
25-JUN-25

         14 Gayathri R
gayu@gmail.com                                                                                       8964785690
Salem
17-AUG-25

         13 Akash Kumar
akash@gmail.com                                                                                      9078237684
Chennai
16-JUL-25

         11 Ravi Chandran
ravi@gmail.com                                                                                       47589364470
Pondicherry
20-MAY-25


20 rows selected.
```

Find customers without any purchases.



```
SQL> SELECT C.* FROM CUSTOMER C LEFT JOIN ORDERS O ON C.CUSTOMER_ID=O.CUSTOMER_ID WHERE O.ORDER_ID IS NULL ORDER BY C.CUSTOMER_ID;

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL                                                                                                PHONE
---------------------------------------------------------------------------------------------------- ---------------
ADDRESS
------------------------------------------------------------------------------------------------------------------------
REGISTRAT
---------
         11 Ravi Chandran
ravi@gmail.com                                                                                       47589364470
Pondicherry
20-MAY-25

         12 Suresh Babu
suresh@gmail.com                                                                                     7895634786
Erode
25-JUN-25

         13 Akash Kumar
akash@gmail.com                                                                                      9078237684
Chennai
16-JUL-25

         14 Gayathri R
gayu@gmail.com                                                                                       8964785690
Salem
17-AUG-25

         15 Ishitha R
ishitha@gmail.com                                                                                    9085682356
Vellore
10-SEP-25
```

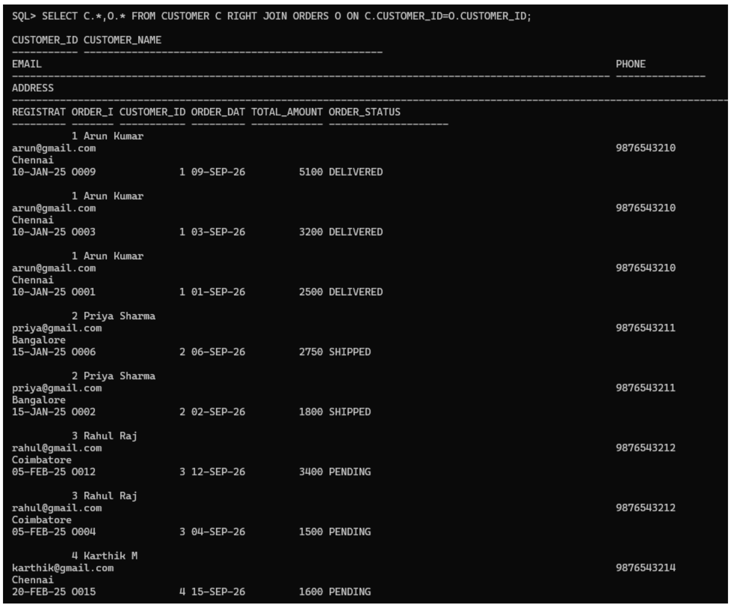
Display all products including products with no sales



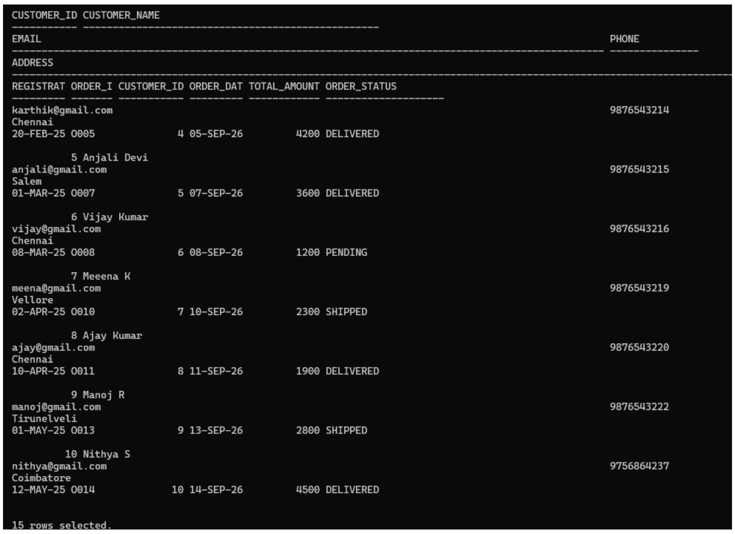
```
SQL> SELECT P.* FROM PRODUCT P LEFT JOIN ORDER_DETAILS OD ON P.PRODUCT_ID=OD.PRODUCT_ID WHERE OD.P
RODUCT_ID IS NULL ORDER BY P.PRODUCT_ID;

no rows selected
```

3. Implement RIGHT JOIN

Display all orders with customer information.

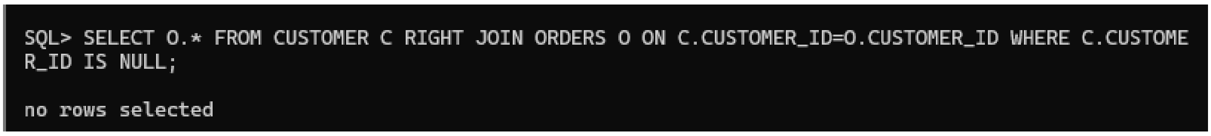


```
SQL> SELECT C.*,O.* FROM CUSTOMER C RIGHT JOIN ORDERS O ON C.CUSTOMER_ID=O.CUSTOMER_ID;

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL                                                                                                PHONE
---------------------------------------------------------------------------------------------------- ---------------
ADDRESS
----------------------------------------------------------------------------------------------------------------------------
REGISTRAT ORDER_I CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
--------- ------- ----------- --------- ------------ --------------------
          1 Arun Kumar
arun@gmail.com                                                                                       9876543210
Chennai
10-JAN-25 O009              1 09-SEP-26         5100 DELIVERED

          1 Arun Kumar
arun@gmail.com                                                                                       9876543210
Chennai
10-JAN-25 O003              1 03-SEP-26         3200 DELIVERED

          1 Arun Kumar
arun@gmail.com                                                                                       9876543210
Chennai
10-JAN-25 O001              1 01-SEP-26         2500 DELIVERED

          2 Priya Sharma
priya@gmail.com                                                                                      9876543211
Bangalore
15-JAN-25 O006              2 06-SEP-26         2750 SHIPPED

          2 Priya Sharma
priya@gmail.com                                                                                      9876543211
Bangalore
15-JAN-25 O002              2 02-SEP-26         1800 SHIPPED

          3 Rahul Raj
rahul@gmail.com                                                                                      9876543212
Coimbatore
05-FEB-25 O012              3 12-SEP-26         3400 PENDING

          3 Rahul Raj
rahul@gmail.com                                                                                      9876543212
Coimbatore
05-FEB-25 O004              3 04-SEP-26         1500 PENDING

          4 Karthik M
karthik@gmail.com                                                                                    9876543214
Chennai
20-FEB-25 O015              4 15-SEP-26         1600 PENDING
```



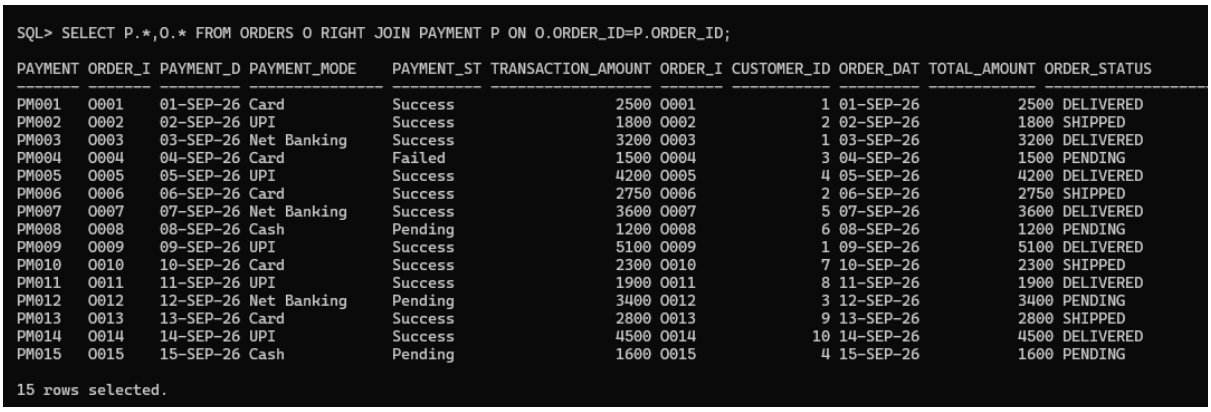
```
CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL                                                                                                PHONE
---------------------------------------------------------------------------------------------------- ---------------
ADDRESS
----------------------------------------------------------------------------------------------------------------------------
REGISTRAT ORDER_I CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
--------- ------- ----------- --------- ------------ --------------------
karthik@gmail.com                                                                                    9876543214
Chennai
20-FEB-25 O005              4 05-SEP-26         4200 DELIVERED

          5 Anjali Devi
anjali@gmail.com                                                                                     9876543215
Salem
01-MAR-25 O007              5 07-SEP-26         3600 DELIVERED

          6 Vijay Kumar
vijay@gmail.com                                                                                      9876543216
Chennai
08-MAR-25 O008              6 08-SEP-26         1200 PENDING

          7 Meeena K
meena@gmail.com                                                                                      9876543219
Vellore
02-APR-25 O010              7 10-SEP-26         2300 SHIPPED

          8 Ajay Kumar
ajay@gmail.com                                                                                       9876543220
Chennai
10-APR-25 O011              8 11-SEP-26         1900 DELIVERED

          9 Manoj R
manoj@gmail.com                                                                                      9876543222
Tirunelveli
01-MAY-25 O013              9 13-SEP-26         2800 SHIPPED

         10 Nithya S
nithya@gmail.com                                                                                     9756864237
Coimbatore
12-MAY-25 O014             10 14-SEP-26         4500 DELIVERED


15 rows selected.
```

Find orders where customer details are missing.



```
SQL> SELECT O.* FROM CUSTOMER C RIGHT JOIN ORDERS O ON C.CUSTOMER_ID=O.CUSTOMER_ID WHERE C.CUSTOME
R_ID IS NULL;

no rows selected
```

Display all payment records with order details



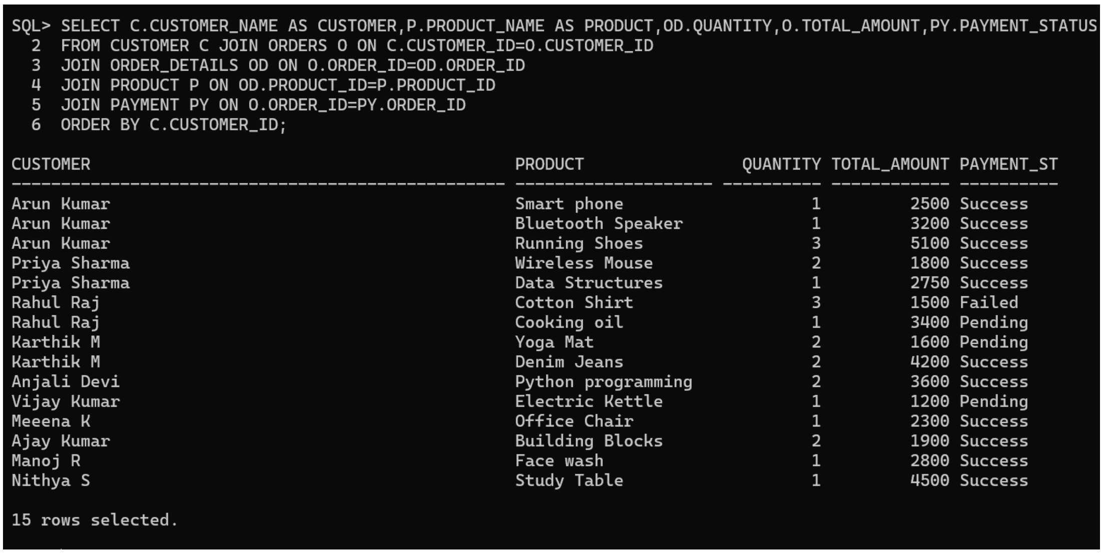
```
SQL> SELECT P.*,O.* FROM ORDERS O RIGHT JOIN PAYMENT P ON O.ORDER_ID=P.ORDER_ID;

PAYMENT ORDER_I PAYMENT_D PAYMENT_MODE    PAYMENT_ST TRANSACTION_AMOUNT ORDER_I CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT ORDER_STATUS
------- ------- --------- --------------- ---------- ------------------ ------- ----------- --------- ------------ -------------------
PM001   O001    01-SEP-26 Card            Success                  2500 O001              1 01-SEP-26         2500 DELIVERED
PM002   O002    02-SEP-26 UPI             Success                  1800 O002              2 02-SEP-26         1800 SHIPPED
PM003   O003    03-SEP-26 Net Banking     Success                  3200 O003              1 03-SEP-26         3200 DELIVERED
PM004   O004    04-SEP-26 Card            Failed                   1500 O004              3 04-SEP-26         1500 PENDING
PM005   O005    05-SEP-26 UPI             Success                  4200 O005              4 05-SEP-26         4200 DELIVERED
PM006   O006    06-SEP-26 Card            Success                  2750 O006              2 06-SEP-26         2750 SHIPPED
PM007   O007    07-SEP-26 Net Banking     Success                  3600 O007              5 07-SEP-26         3600 DELIVERED
PM008   O008    08-SEP-26 Cash            Pending                  1200 O008              6 08-SEP-26         1200 PENDING
PM009   O009    09-SEP-26 UPI             Success                  5100 O009              1 09-SEP-26         5100 DELIVERED
PM010   O010    10-SEP-26 Card            Success                  2300 O010              7 10-SEP-26         2300 SHIPPED
PM011   O011    11-SEP-26 UPI             Success                  1900 O011              8 11-SEP-26         1900 DELIVERED
PM012   O012    12-SEP-26 Net Banking     Pending                  3400 O012              3 12-SEP-26         3400 PENDING
PM013   O013    13-SEP-26 Card            Success                  2800 O013              9 13-SEP-26         2800 SHIPPED
PM014   O014    14-SEP-26 UPI             Success                  4500 O014             10 14-SEP-26         4500 DELIVERED
PM015   O015    15-SEP-26 Cash            Pending                  1600 O015              4 15-SEP-26         1600 PENDING

15 rows selected.
```

Retrieve Complete Order Details

Task:

Generate a complete order report by combining:

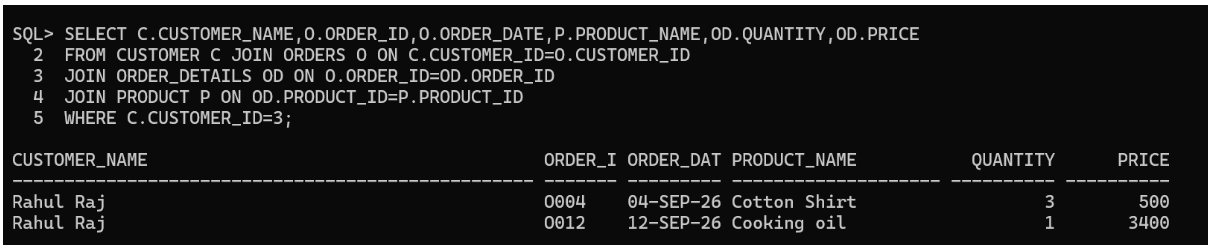
- Customer
- Name
- Product Name
- Quantity
- Order Date
- Total Amount
- Payment Status



```
SQL> SELECT C.CUSTOMER_NAME AS CUSTOMER,P.PRODUCT_NAME AS PRODUCT,OD.QUANTITY,O.TOTAL_AMOUNT,PY.PAYMENT_STATUS
  2  FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID=O.CUSTOMER_ID
  3  JOIN ORDER_DETAILS OD ON O.ORDER_ID=OD.ORDER_ID
  4  JOIN PRODUCT P ON OD.PRODUCT_ID=P.PRODUCT_ID
  5  JOIN PAYMENT PY ON O.ORDER_ID=PY.ORDER_ID
  6  ORDER BY C.CUSTOMER_ID;

CUSTOMER                                           PRODUCT                QUANTITY TOTAL_AMOUNT PAYMENT_ST
-------------------------------------------------- -------------------- ---------- ------------ ----------
Arun Kumar                                         Smart phone                   1         2500 Success
Arun Kumar                                         Bluetooth Speaker             1         3200 Success
Arun Kumar                                         Running Shoes                 3         5100 Success
Priya Sharma                                       Wireless Mouse                2         1800 Success
Priya Sharma                                       Data Structures               1         2750 Success
Rahul Raj                                          Cotton Shirt                  3         1500 Failed
Rahul Raj                                          Cooking oil                   1         3400 Pending
Karthik M                                          Yoga Mat                      2         1600 Pending
Karthik M                                          Denim Jeans                   2         4200 Success
Anjali Devi                                        Python programming            2         3600 Success
Vijay Kumar                                        Electric Kettle               1         1200 Pending
Meeena K                                           Office Chair                  1         2300 Success
Ajay Kumar                                         Building Blocks               2         1900 Success
Manoj R                                            Face wash                     1         2800 Success
Nithya S                                           Study Table                   1         4500 Success

15 rows selected.
```

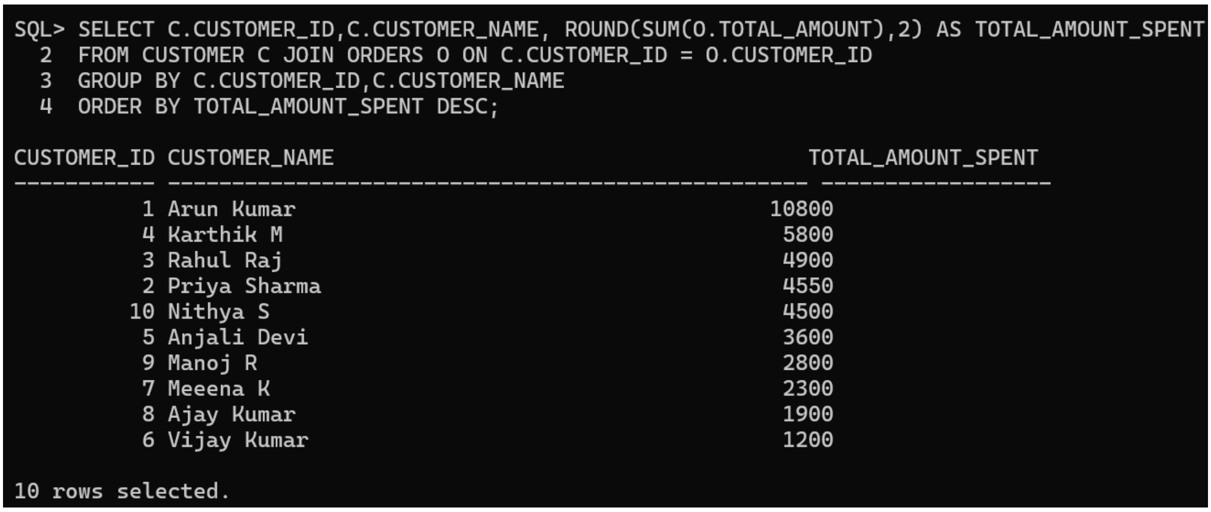
Display Customer Purchase History

All products purchased by a customer.



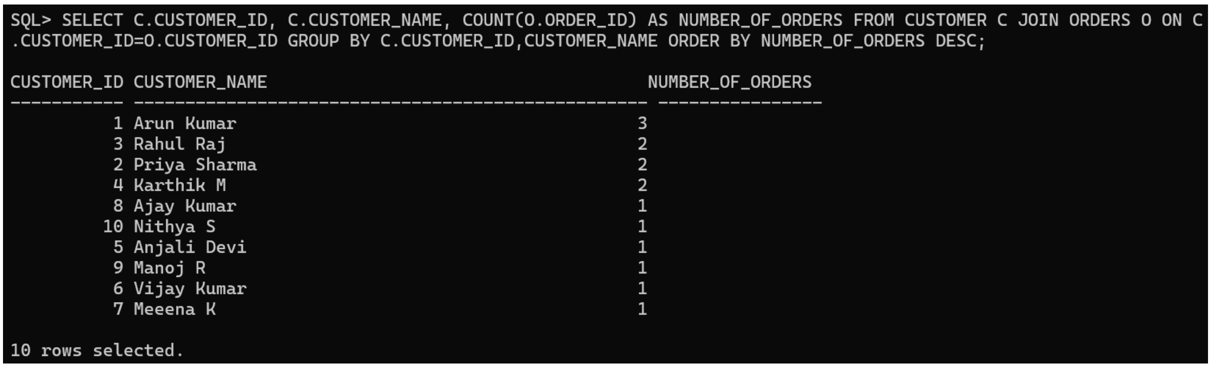
```
SQL> SELECT C.CUSTOMER_NAME,O.ORDER_ID,O.ORDER_DATE,P.PRODUCT_NAME,OD.QUANTITY,OD.PRICE
  2  FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID=O.CUSTOMER_ID
  3  JOIN ORDER_DETAILS OD ON O.ORDER_ID=OD.ORDER_ID
  4  JOIN PRODUCT P ON OD.PRODUCT_ID=P.PRODUCT_ID
  5  WHERE C.CUSTOMER_ID=3;

CUSTOMER_NAME                                      ORDER_I ORDER_DAT PRODUCT_NAME           QUANTITY      PRICE
-------------------------------------------------- ------- --------- -------------------- ---------- ----------
Rahul Raj                                          O004    04-SEP-26 Cotton Shirt                  3        500
Rahul Raj                                          O012    12-SEP-26 Cooking oil                   1       3400
```

Total amount spent by each customer



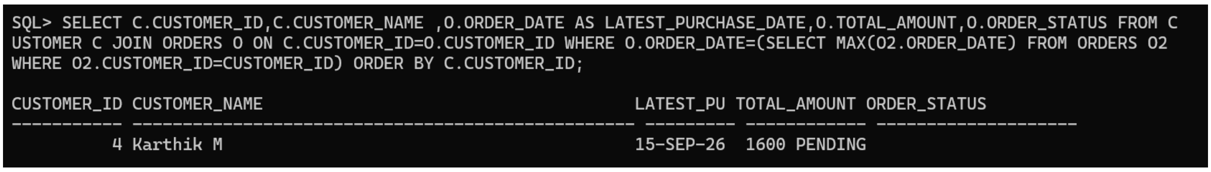
```
SQL> SELECT C.CUSTOMER_ID,C.CUSTOMER_NAME, ROUND(SUM(O.TOTAL_AMOUNT),2) AS TOTAL_AMOUNT_SPENT
  2  FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID = O.CUSTOMER_ID
  3  GROUP BY C.CUSTOMER_ID,C.CUSTOMER_NAME
  4  ORDER BY TOTAL_AMOUNT_SPENT DESC;

CUSTOMER_ID CUSTOMER_NAME                                      TOTAL_AMOUNT_SPENT
----------- -------------------------------------------------- ------------------
          1 Arun Kumar                                     10800
          4 Karthik M                                       5800
          3 Rahul Raj                                       4900
          2 Priya Sharma                                    4550
         10 Nithya S                                        4500
          5 Anjali Devi                                     3600
          9 Manoj R                                         2800
          7 Meeena K                                        2300
          8 Ajay Kumar                                      1900
          6 Vijay Kumar                                     1200

10 rows selected.
```

Number of orders placed by each customer



```
SQL> SELECT C.CUSTOMER_ID, C.CUSTOMER_NAME, COUNT(O.ORDER_ID) AS NUMBER_OF_ORDERS FROM CUSTOMER C JOIN ORDERS O ON C
.CUSTOMER_ID=O.CUSTOMER_ID GROUP BY C.CUSTOMER_ID,CUSTOMER_NAME ORDER BY NUMBER_OF_ORDERS DESC;

CUSTOMER_ID CUSTOMER_NAME                                      NUMBER_OF_ORDERS
----------- -------------------------------------------------- ----------------
          1 Arun Kumar                                       3
          3 Rahul Raj                                        2
          2 Priya Sharma                                     2
          4 Karthik M                                        2
          8 Ajay Kumar                                       1
         10 Nithya S                                         1
          5 Anjali Devi                                      1
          9 Manoj R                                          1
          6 Vijay Kumar                                      1
          7 Meeena K                                         1

10 rows selected.
```

Latest purchase details of customers



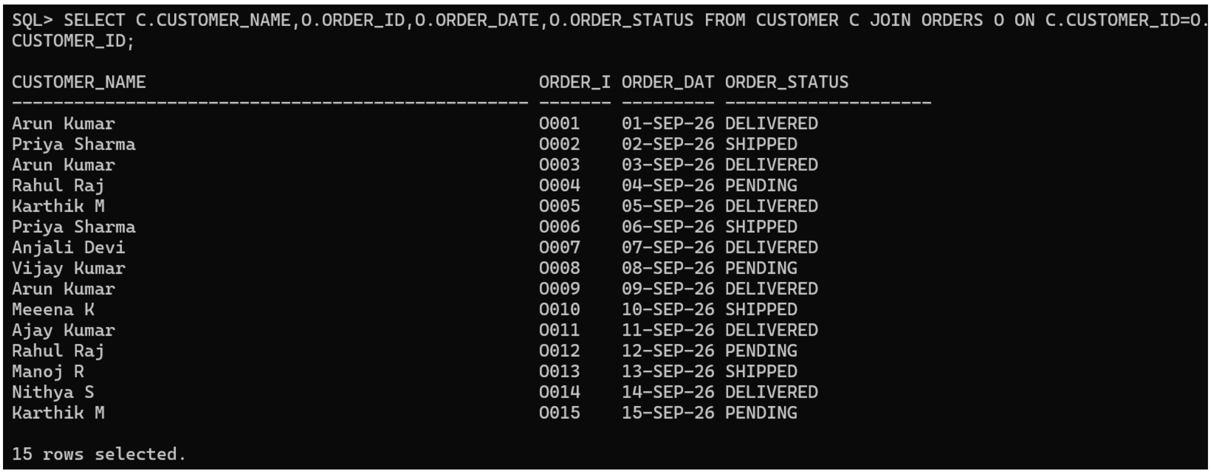
```
SQL> SELECT C.CUSTOMER_ID,C.CUSTOMER_NAME ,O.ORDER_DATE AS LATEST_PURCHASE_DATE,O.TOTAL_AMOUNT,O.ORDER_STATUS FROM C
USTOMER C JOIN ORDERS O ON C.CUSTOMER_ID=O.CUSTOMER_ID WHERE O.ORDER_DATE=(SELECT MAX(O2.ORDER_DATE) FROM ORDERS O2
WHERE O2.CUSTOMER_ID=CUSTOMER_ID) ORDER BY C.CUSTOMER_ID;

CUSTOMER_ID CUSTOMER_NAME                                      LATEST_PU TOTAL_AMOUNT ORDER_STATUS
----------- -------------------------------------------------- --------- ------------ --------------------
          4 Karthik M                                          15-SEP-26  1600 PENDING
```

Generate Multi-Table Business Reports

Report 1: Customer Order Report

Display:

- Customer name
- Order ID
- Order date
- Order status

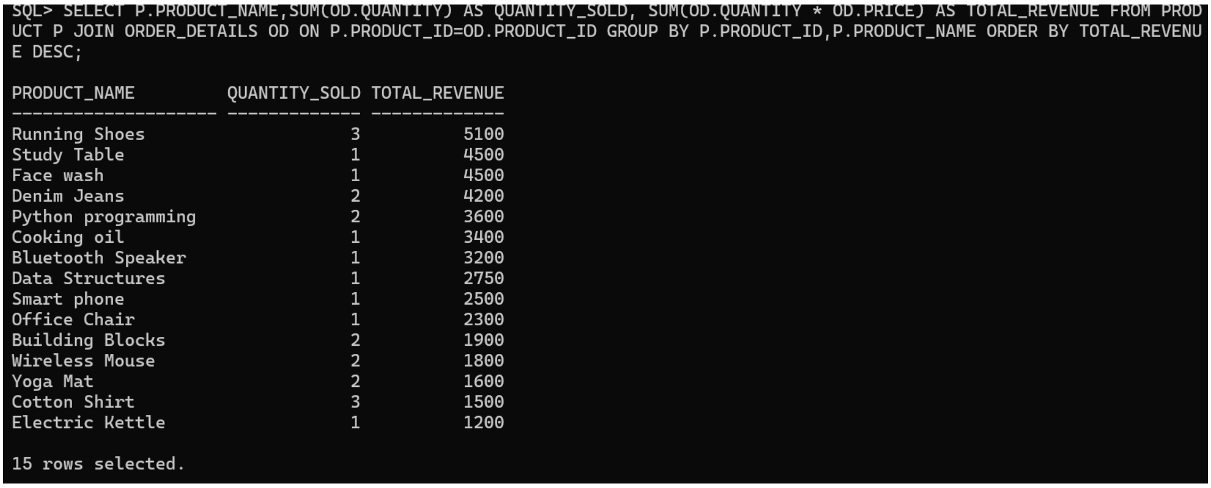


```
SQL> SELECT C.CUSTOMER_NAME,O.ORDER_ID,O.ORDER_DATE,O.ORDER_STATUS FROM CUSTOMER C JOIN ORDERS O ON C.CUSTOMER_ID=O.
CUSTOMER_ID;

CUSTOMER_NAME                                      ORDER_I ORDER_DAT ORDER_STATUS
-------------------------------------------------- ------- --------- --------------------
Arun Kumar                                         O001    01-SEP-26 DELIVERED
Priya Sharma                                       O002    02-SEP-26 SHIPPED
Arun Kumar                                         O003    03-SEP-26 DELIVERED
Rahul Raj                                          O004    04-SEP-26 PENDING
Karthik M                                          O005    05-SEP-26 DELIVERED
Priya Sharma                                       O006    06-SEP-26 SHIPPED
Anjali Devi                                        O007    07-SEP-26 DELIVERED
Vijay Kumar                                        O008    08-SEP-26 PENDING
Arun Kumar                                         O009    09-SEP-26 DELIVERED
Meeena K                                           O010    10-SEP-26 SHIPPED
Ajay Kumar                                         O011    11-SEP-26 DELIVERED
Rahul Raj                                          O012    12-SEP-26 PENDING
Manoj R                                            O013    13-SEP-26 SHIPPED
Nithya S                                           O014    14-SEP-26 DELIVERED
Karthik M                                          O015    15-SEP-26 PENDING

15 rows selected.
```

Report 2: Sales Report

Display:

- Product name
- Quantity sold
- Total revenue

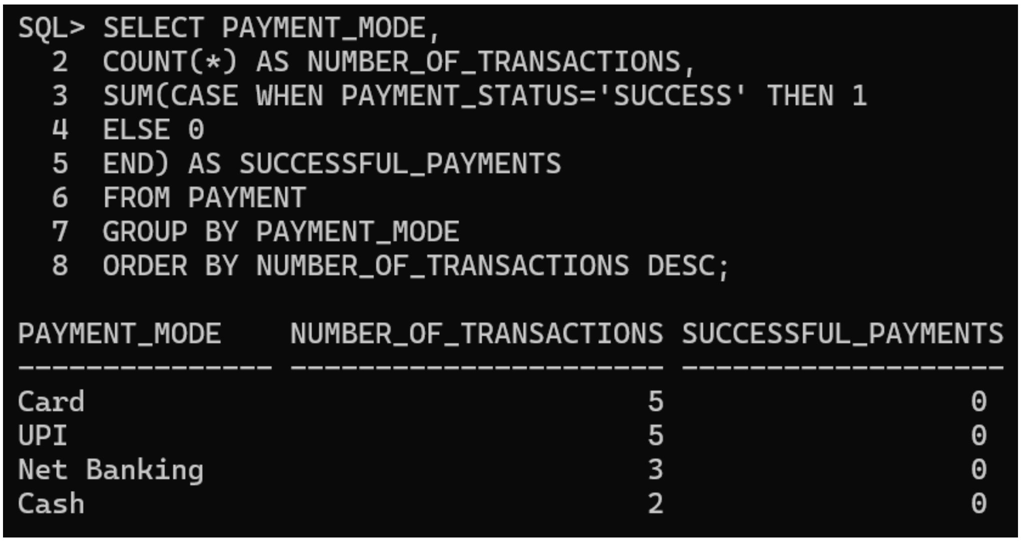


```
SQL> SELECT P.PRODUCT_NAME,SUM(OD.QUANTITY) AS QUANTITY_SOLD, SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE FROM PROD
UCT P JOIN ORDER_DETAILS OD ON P.PRODUCT_ID=OD.PRODUCT_ID GROUP BY P.PRODUCT_ID,P.PRODUCT_NAME ORDER BY TOTAL_REVENU
E DESC;

PRODUCT_NAME         QUANTITY_SOLD TOTAL_REVENUE
-------------------- ------------- -------------
Running Shoes                    3          5100
Study Table                      1          4500
Face wash                        1          4500
Denim Jeans                      2          4200
Python programming               2          3600
Cooking oil                      1          3400
Bluetooth Speaker                1          3200
Data Structures                  1          2750
Smart phone                      1          2500
Office Chair                     1          2300
Building Blocks                  2          1900
Wireless Mouse                   2          1800
Yoga Mat                         2          1600
Cotton Shirt                     3          1500
Electric Kettle                  1          1200

15 rows selected.
```

Report 3: Payment Analysis Report

Display:

- Payment mode
- Number of transactions
- Successful payments

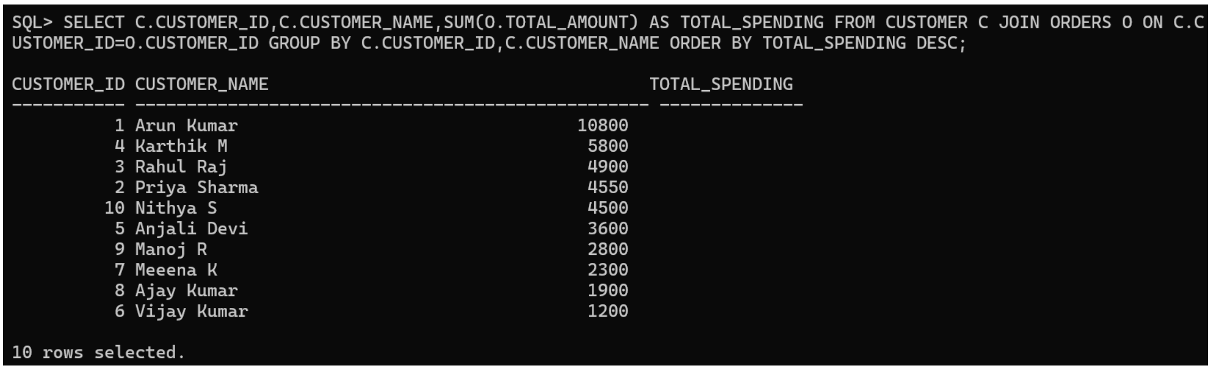


```
SQL> SELECT PAYMENT_MODE,
  2  COUNT(*) AS NUMBER_OF_TRANSACTIONS,
  3  SUM(CASE WHEN PAYMENT_STATUS='SUCCESS' THEN 1
  4  ELSE 0
  5  END) AS SUCCESSFUL_PAYMENTS
  6  FROM PAYMENT
  7  GROUP BY PAYMENT_MODE
  8  ORDER BY NUMBER_OF_TRANSACTIONS DESC;

PAYMENT_MODE    NUMBER_OF_TRANSACTIONS SUCCESSFUL_PAYMENTS
--------------- ---------------------- -------------------
Card                                 5                   0
UPI                                  5                   0
Net Banking                          3                   0
Cash                                 2                   0
```

Report 4: Customer Purchase Analysis

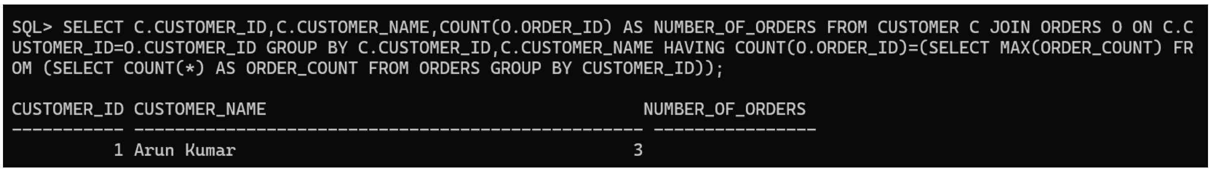
Find:

- Top purchasing customers



```
SQL> SELECT C.CUSTOMER_ID,C.CUSTOMER_NAME,SUM(O.TOTAL_AMOUNT) AS TOTAL_SPENDING FROM CUSTOMER C JOIN ORDERS O ON C.C
USTOMER_ID=O.CUSTOMER_ID GROUP BY C.CUSTOMER_ID,C.CUSTOMER_NAME ORDER BY TOTAL_SPENDING DESC;

CUSTOMER_ID CUSTOMER_NAME                                     TOTAL_SPENDING
----------- -------------------------------------------------- --------------
          1 Arun Kumar                                 10800
          4 Karthik M                                   5800
          3 Rahul Raj                                   4900
          2 Priya Sharma                                4550
         10 Nithya S                                    4500
          5 Anjali Devi                                 3600
          9 Manoj R                                     2800
          7 Meeena K                                    2300
          8 Ajay Kumar                                  1900
          6 Vijay Kumar                                 1200

10 rows selected.
```

- Customers with maximum orders



```
SQL> SELECT C.CUSTOMER_ID,C.CUSTOMER_NAME,COUNT(O.ORDER_ID) AS NUMBER_OF_ORDERS FROM CUSTOMER C JOIN ORDERS O ON C.C
USTOMER_ID=O.CUSTOMER_ID GROUP BY C.CUSTOMER_ID,C.CUSTOMER_NAME HAVING COUNT(O.ORDER_ID)=(SELECT MAX(ORDER_COUNT) FR
OM (SELECT COUNT(*) AS ORDER_COUNT FROM ORDERS GROUP BY CUSTOMER_ID));

CUSTOMER_ID CUSTOMER_NAME                                     NUMBER_OF_ORDERS
----------- -------------------------------------------------- ----------------
          1 Arun Kumar                                       3
```